HK IMO Training 2019-2020
Problem Set 8
For 21 December 2019

Problem 29. Consider $\triangle ABC$ with $BC = a$, $CA = b$, $AB = c$, and circumradius R. Let X, Y, Z be points on the interior of sides BC, CA and AB, respectively. Prove that

$$(XY + YZ + ZX)\left(\frac{1}{AX} + \frac{1}{BY} + \frac{1}{CZ}\right) \geqslant \frac{a+b+c}{R}.$$

Problem 30. Consider $\triangle ABC$ with $BC = a$, $CA = b$, and $AB = c$. Prove that

$$\frac{2bc\cos A}{b+c} < b + c - a < \frac{2bc}{a}.$$

Problem 31. Let G, R and r be, respectively, the centroid, the circumradius and inradius of $\triangle ABC$. Prove that

$$2r \leqslant \frac{AG + BG + CG}{3} \leqslant R.$$

Problem 32. Given $\triangle ABC$, prove that

$$\sin A \sin B \sin C\left(\frac{1}{\sin A + \sin B} + \frac{1}{\sin B + \sin C} + \frac{1}{\sin C + \sin A}\right) \leqslant \frac{3}{4}(\cos A + \cos B + \cos C).$$